DATA ANALYTICS & BUSINESS INTELLIGENCE | CAPSTONE

Executive Decision Report

From profitable growth to growth that pays: three decisions to lift annual profit by about $97k

Prepared for: Executive Leadership Team

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Scope: 3 years of omnichannel retail orders (2023-2025), 11,996 orders, 2,116 customers

EXECUTIVE SUMMARY

Revenue grew 30% in 2025, but margin fell to 11.9%. Three decisions recover about $97k a year

$1.40M
2025 revenue, +30% YoY

$167k
2025 profit, only +9.5% YoY

11.9%
profit margin, down from 15.4% in 2023

+58%
profit uplift at run-rate ($97k / yr, Base case)

What we found

- **Discounts above 20% lose money.** Margin is -0.6% at 21-30% off and -22.6% above 30%. In 2025, 32% of orders sat in this zone and earned -$40k.
- **Marketplace is the fastest-growing and the only loss-making channel** (-2.5% margin). Its revenue share rose from 12% to 26%.
- **Late delivery triples returns** (14.9% vs 4.8%). Central region ships 58% of orders late.

What we recommend

1. **Discount guardrails:** cap at 20%, approval above. **+$63k / yr**
2. **Marketplace margin repair:** 10% cap and fee renegotiation. **+$18k / yr**
3. **Delivery SLA and retention programme.** **+$17k / yr**

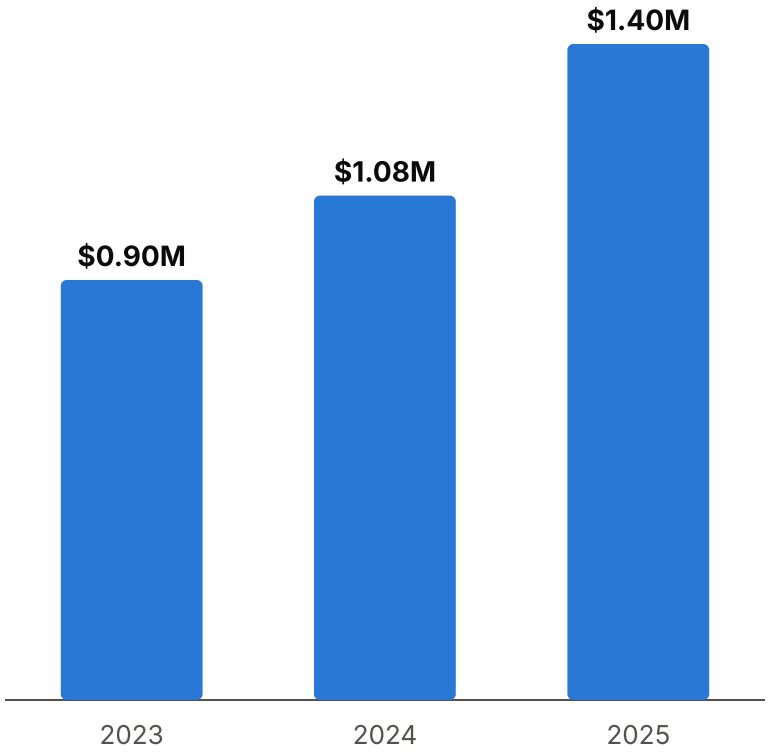
Investment: $85k over 3 years. **Base-case net benefit:** $176k (208% ROI), payback in month 10. Revenue is traded down by 8% because loss-making sales are removed.

Source: omnichannel retail orders 2023-25 (simulated dataset, 11,996 clean order lines). Analysis: notebook Executive_Decision_Analysis.ipynb.

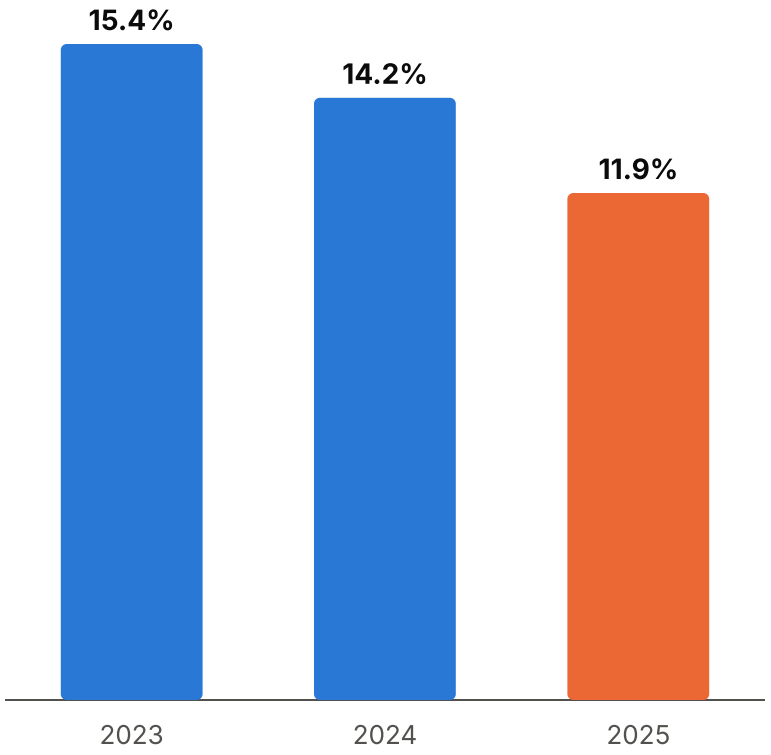
PERFORMANCE SNAPSHOT

Growth is real, but profit is not keeping pace: revenue +56% since 2023, profit +21%

Revenue ($M)



Profit margin (%)



So what

- Orders rose from 3,079 to 4,978 and active customers from 524 to 1,376.
- Average order value is flat ($281), so growth comes from volume, not value.
- Margin has lost 3.5 points in two years. Each extra revenue dollar now earns less profit.
- Return rate (7.4%) and late shipments (23%) have not improved.

Source: omnichannel retail orders 2023-25 (simulated dataset, 11,996 clean order lines). Analysis: notebook Executive_Decision_Analysis.ipynb.

METHODOLOGY

Method: a reproducible five-step pipeline from raw orders to a costed decision

1 Ingest

12,400 raw order lines, 3 date formats, text-formatted money fields

2 Clean

400 duplicates removed, 1,260 missing cells resolved, labels standardised (region 29 to 5, category 18 to 3), 28 entry errors fixed

3 Analyse

11,996 clean orders. Revenue, margin, channel, customer and delivery drivers measured

4 Test

3 hypotheses tested at 5% significance (Welch t-test, Mann-Whitney U, chi-square)

5 Model

Order-level ROI model on 2025 baseline, 3 scenarios, 36-month cash flow

Quality controls

- Sales arithmetic reconciles on 100% of rows after cleaning.
- Discounts and quantities recovered from each order's own arithmetic before any outlier fix.
- Only 4 rows dropped as unrecoverable; 300 missing ship times imputed and flagged.
- Every figure in this deck is generated from the notebook's exported metrics.

Scope and limits

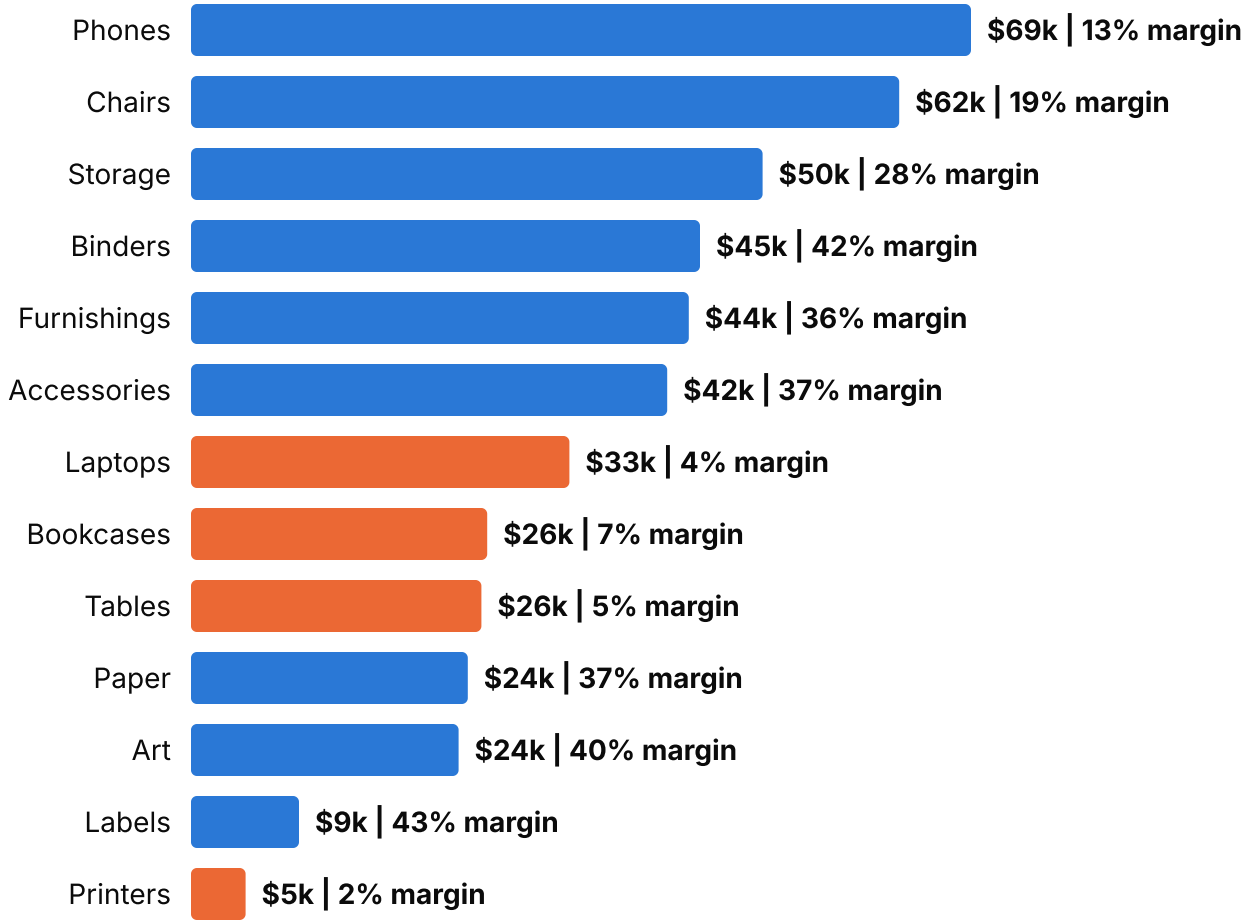
- The dataset is simulated (seeded, reproducible) to mirror an omnichannel retailer; findings show the method and decision logic.
- ROI is a model, not a forecast. Behavioural inputs (retention after a discount cap, fee cuts, late-delivery reduction) are assumptions to validate in pilots.
- Benefits are held at 2025 volumes. No market growth is assumed.

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KEY REVENUE DRIVERS

Revenue driver 1: Office Supplies earns a third of profit on 13% of revenue; Technology and Furniture are volume with thin margins

Profit by sub-category, 2023-25 (orange = margin below 10%)



So what

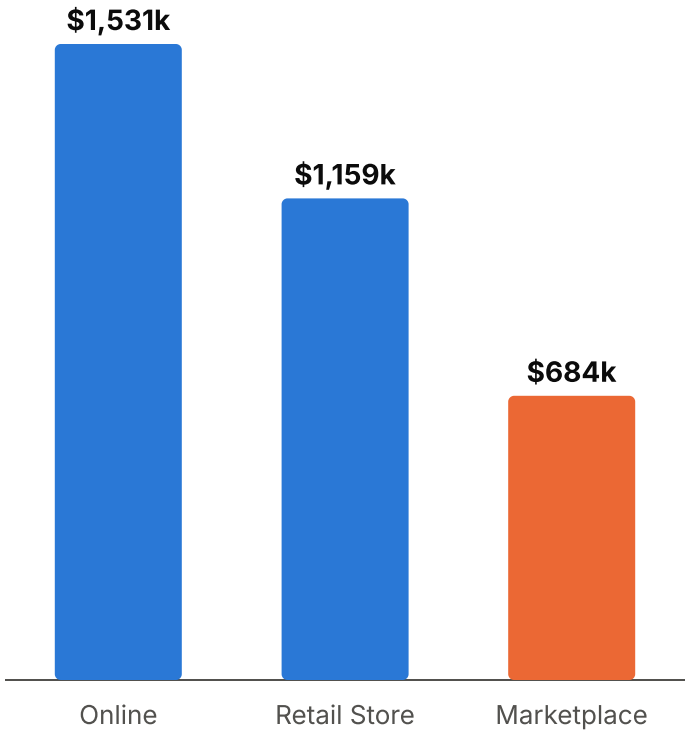
- **Technology**: 49% of revenue, 32% of profit, margin 9.0%.
- **Furniture**: 38% of revenue, margin 12.2%.
- **Office Supplies**: 13% of revenue but 33% of profit at 35% margin.
- Laptops, Tables and Bookcases add large revenue at margins under 8% (Printers are lower still, on small volume). They are the first candidates for price and discount discipline.

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KEY REVENUE DRIVERS

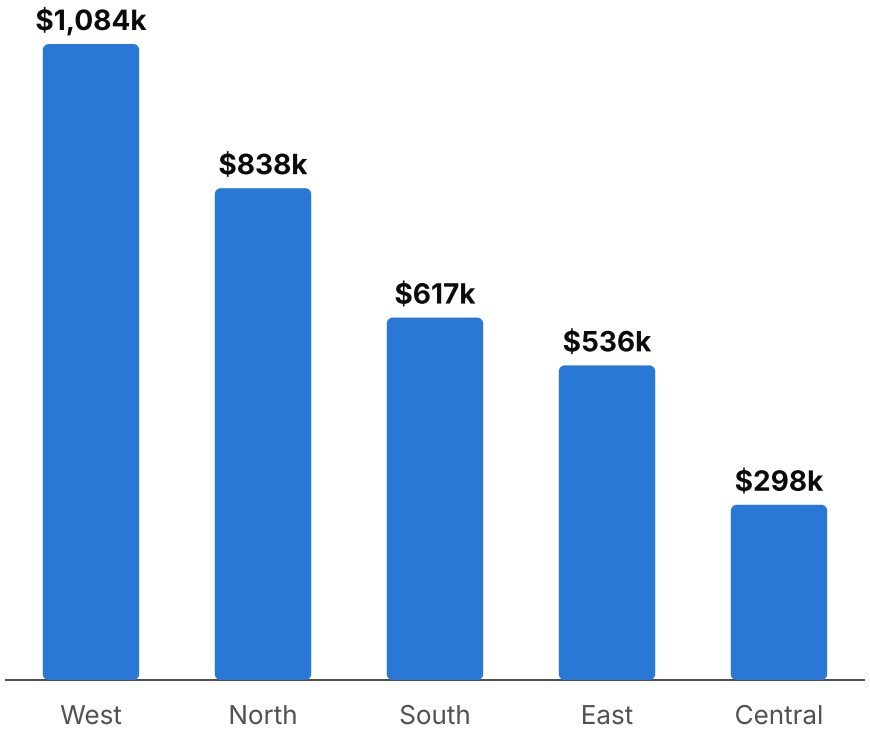
Revenue driver 2: Online and Retail carry the profit; the top 20% of customers deliver two thirds of revenue

Revenue by channel ($k)



Orange = loss-making channel

Revenue by region ($k)



Margin is similar everywhere (13.2-13.9%)

66%
of revenue from top 20% of 2,116 customers

50%
customers retained 2024 to 2025 (was 54%)

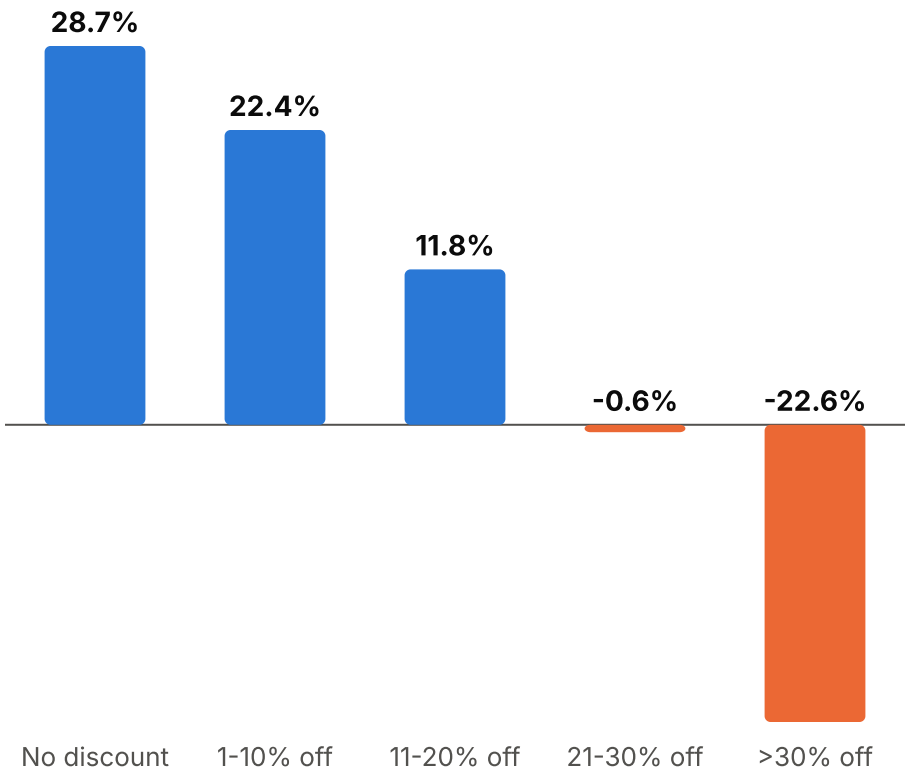
71%
of customers order more than once

Source: omnichannel retail orders 2023-25 (simulated dataset, 11,996 clean order lines). Analysis: notebook Executive_Decision_Analysis.ipynb.

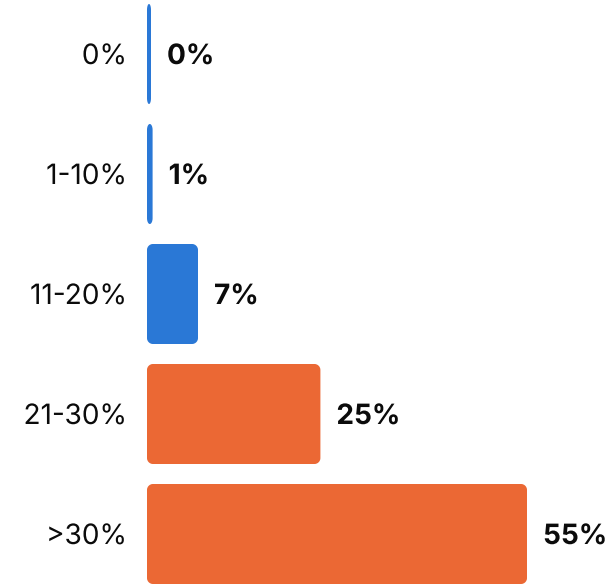
RISK AREAS

Risk 1: discounts above 20% turn profitable orders into losses (correlation with margin -0.70)

Profit margin by discount band



Share of orders that lose money



Loss-making = negative order profit

So what

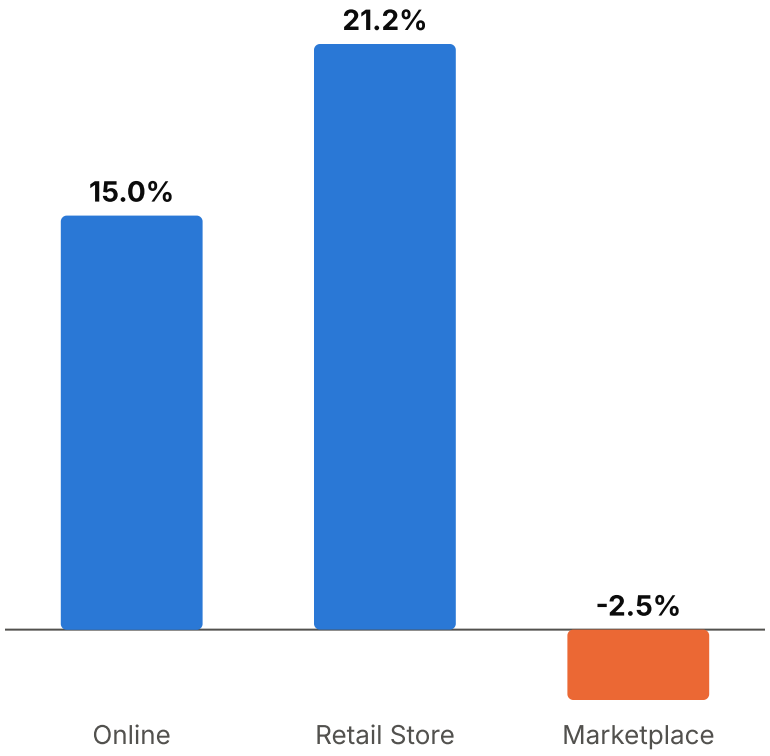
- 1,605 orders in 2025 (32%) were discounted above 20%. They brought $389k of revenue and -$40k of profit.
- 1,579 loss-making orders (13% of all orders) destroyed $129k.
- Every band at or below 20% earns a positive margin.

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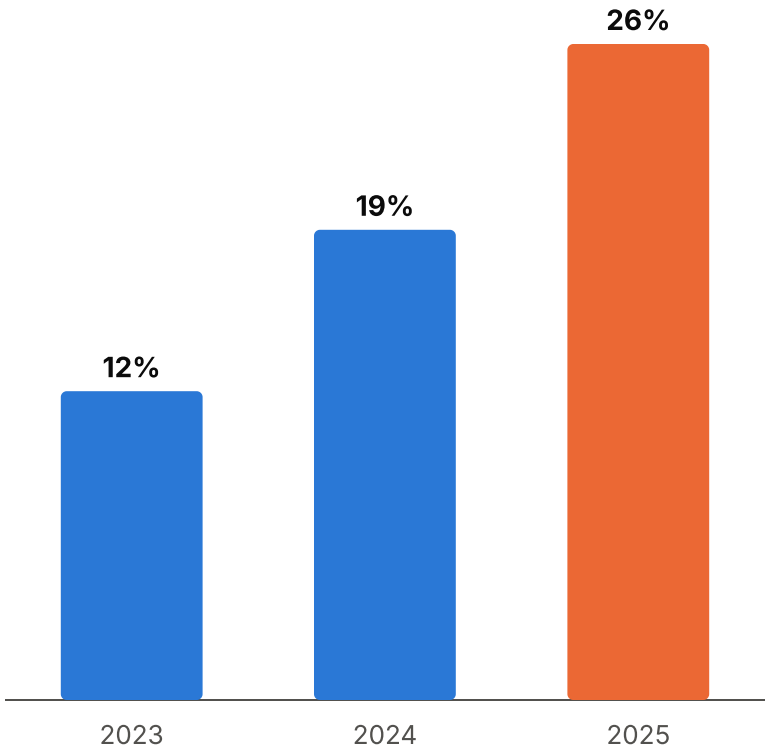
RISK AREAS

Risk 2: Marketplace is the fastest-growing channel and the only one that loses money

Profit margin by channel



Marketplace share of revenue



So what

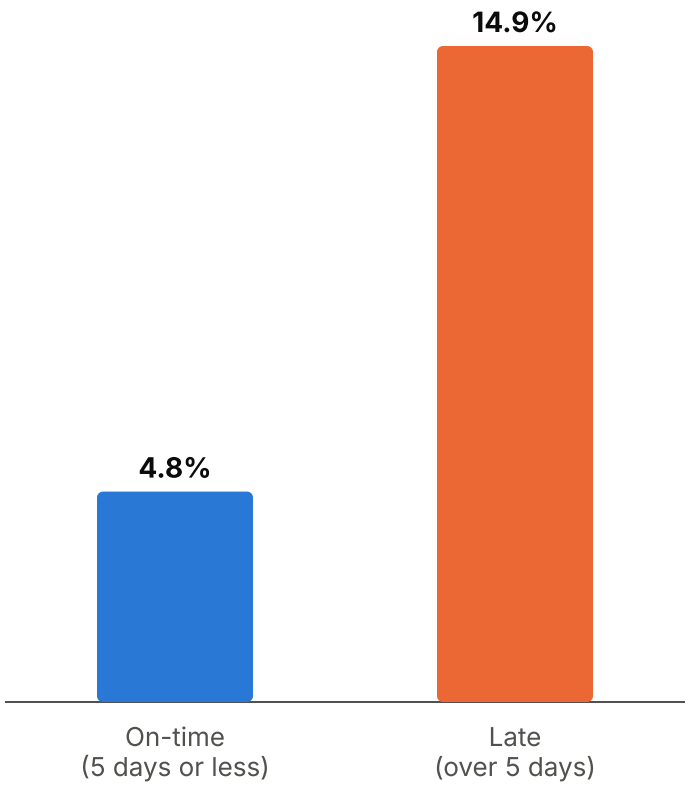
- Marketplace lost $17k on $684k of revenue (fees of 13% plus deep discounts).
- Its share doubled in two years. Each point of mix shift from Retail to Marketplace lowers company margin.
- Marketplace orders earn 9.6% average margin vs 29.6% elsewhere (H2).

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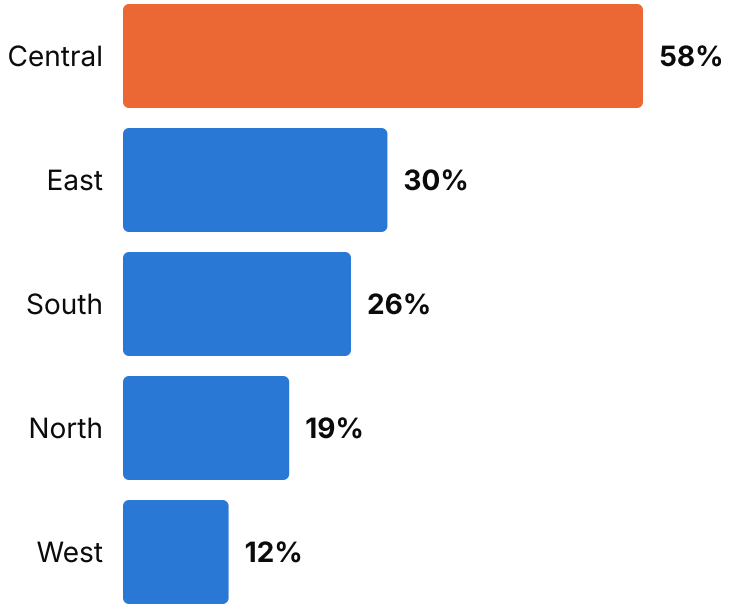
RISK AREAS

Risk 3: late deliveries are returned 3 times as often, and Central ships 58% of orders late

Return rate by delivery speed



Late shipments by region (% of orders)



So what

- Late orders are 2,730 of 11,996 (23%). Each return costs the sale plus handling.
- Central and East account for the worst service. Central alone is 4.9x West.
- Weak delivery also feeds the falling retention rate (50%).

Source: omnichannel retail orders 2023-25 (simulated dataset, 11,996 clean order lines). Analysis: notebook Executive_Decision_Analysis.ipynb.

HYPOTHESIS TESTING

All three business hypotheses are statistically confirmed, so the recommendations rest on evidence, not anecdote

	Hypothesis	Observed	Test	Significance	Verdict
H1	Orders discounted over 20% earn a lower margin	4.9% vs 33.6%	Welch t-test + Mann-Whitney U	p < 0.001	**Supported**
H2	Marketplace orders earn a lower margin than other channels	9.6% vs 29.6%	Welch t-test + Mann-Whitney U	p < 0.001	**Supported**
H3	Late shipments are returned more often	14.9% vs 4.8% return rate	Chi-square test of independence	p < 0.001	**Supported**

How to read this

- Sample sizes are large: 3,511 vs 8,485 orders (H1), 2,660 vs 9,336 (H2), 2,730 vs 9,266 (H3).
- Parametric and non-parametric tests agree for H1 and H2.

Caveat

- The tests show association in this data. Pilots (see roadmap) confirm cause and effect before full rollout, especially on how customers react to a discount cap.

Source: omnichannel retail orders 2023-25 (simulated dataset, 11,996 clean order lines). Analysis: notebook Executive_Decision_Analysis.ipynb.

STRATEGIC RECOMMENDATIONS

Three strategic recommendations, sequenced so benefits are never double counted

Discount guardrails 1

+$63k / yr

Action. Cap discounts at 20%. Anything higher needs manager approval with a margin check.

Mechanism. Re-prices 1,605 loss-prone 2025 orders; assumes 70% of those customers still buy.

KPI. Share of orders above 20% off: 32% to under 5%; loss-making orders under 5%.

Marketplace margin repair 2

+$18k / yr

Action. Renegotiate marketplace fees, cap marketplace discounts at 10%, set price floors on low-margin SKUs.

Mechanism. Fee cut of 1 point and a 10% cap; assumes 70% of affected orders are kept.

KPI. Marketplace margin from -2.5% to positive.

Delivery SLA and retention 3

+$17k / yr

Action. Add a regional fulfilment partner for Central and East. Launch SLA tracking and loyalty tiers.

Mechanism. Cuts late deliveries by 40%, so their return rate falls to the on-time rate, and lifts retention by 5 points.

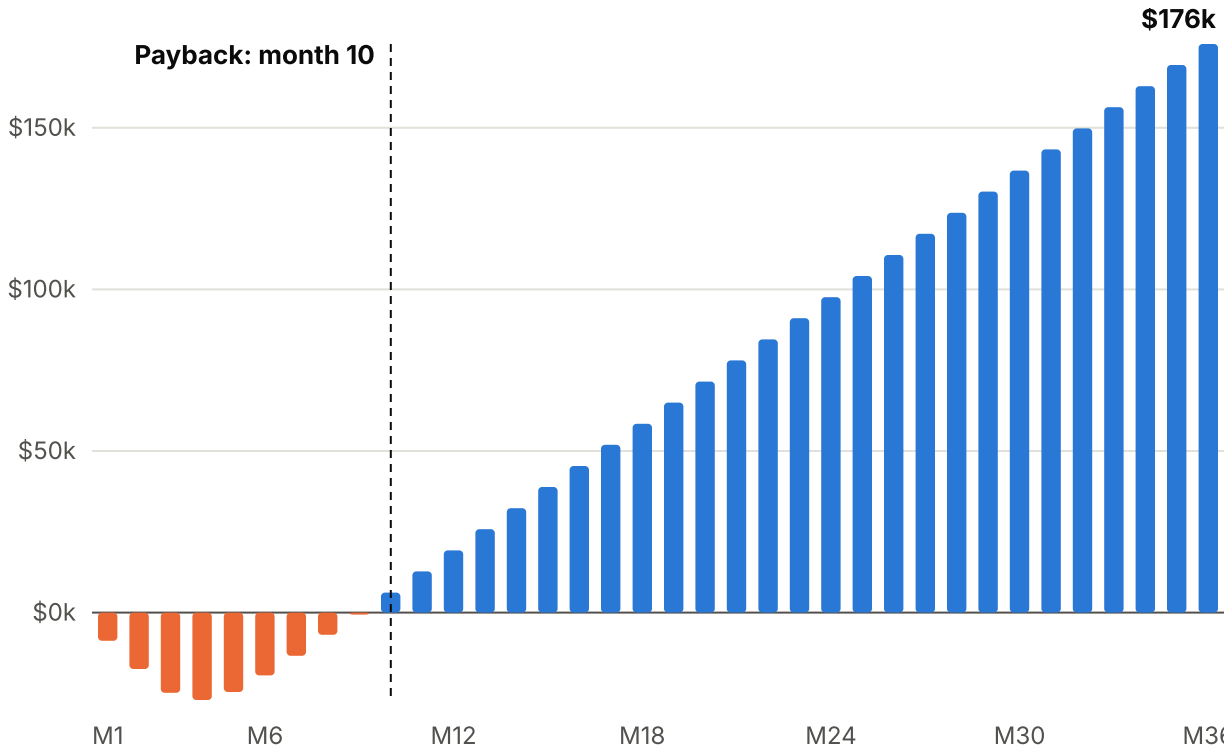
KPI. Late shipments from 23% overall (Central 58%) to under 15%; retention above 55%.

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ROI PROJECTIONS

ROI: $85k invested returns $176k net over 3 years (208% ROI), paying back in month 10

Cumulative net cash flow, Base case



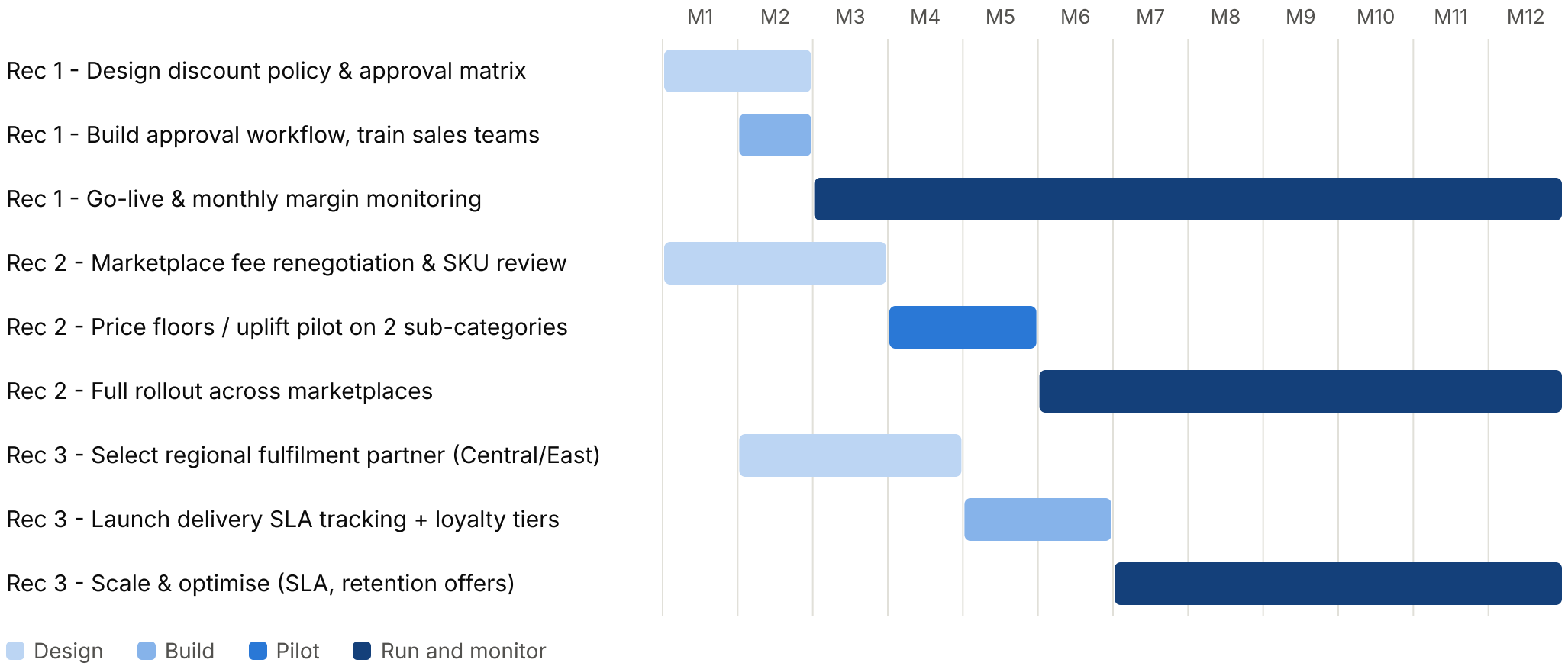
Recommendation	Run-rate / yr	Net: Cons.	Net: Base	Net: Opt.	ROI (Base)	Payback (Base)
1 Discount guardrails	$63k	$138k	**$151k**	$160k	708%	Month 6
2 Marketplace repair	$18k	-$2k	**$18k**	$40k	65%	Month 18
3 SLA + retention	$17k	-$10k	**$7k**	$23k	19%	Month 26
Total	**$97k**	**$126k**	**$176k**	**$223k**	**208%**	**Month 10**

- Investment: $34k one-time plus $19k a year to run ($85k over 36 months). Net = 3-year benefit minus cost.
- Base run-rate profit rises from $167k to $264k; revenue is $113k lower (8%).
- The Conservative case still pays back in month 11. There Recs 2 and 3 are marginal, so Rec 1 carries the case.

Model: 2025 order-level baseline, sequential application, benefits ramp over 3 months from go-live. Parameters are assumptions (notebook, Section 6).

IMPLEMENTATION TIMELINE

Implementation roadmap: 12 months, with Rec 1 live in month 3 and a full run-rate by month 12



Month 3

Discount policy live. First margin report to the executive team.

Month 6

Marketplace pilot readout and Central/East partner signed. Go / no-go for full rollout.

Month 12

All three programmes at run-rate. Review actual vs modelled benefit and reset targets.

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Decisions requested today, and how we will manage the risks

We ask the leadership team to approve

1. **Policy.** A company-wide 20% discount cap with an approval matrix, effective month 3.
2. **Funding.** $34k one-time and $19k a year, released in stages at months 3 and 6.
3. **Ownership.** One executive sponsor per recommendation and a monthly margin review.

Success scorecard: company margin from 11.9% toward 21% (model-implied), marketplace margin above 0%, late shipments under 15%, retention above 55%.

Key risks and mitigations

Risk	Mitigation
Customers leave when discounts are capped	Pilot in two regions first; retention assumed at 70% (Base), tested down to 55%
Marketplaces resist fee changes or delist us	Start with SKU-level price floors; keep the option to reduce exposure on loss-making SKUs
Delivery partner underperforms	SLA-linked contract with penalty clauses; weekly late-delivery dashboard
Model assumptions prove optimistic	Stage-gate funding at months 3 and 6; Conservative case still repays the investment

Source: omnichannel retail orders 2023-25 (simulated dataset, 11,996 clean order lines). Analysis: notebook Executive_Decision_Analysis.ipynb.

APPENDIX

Appendix: analytics portfolio, data quality log and model assumptions

Portfolio deliverables

- **Executive_Decision_Analysis.ipynb**: full analysis, 9 sections, all charts
- **Executive_Decision_Analysis.html**: read-only export of the notebook
- **raw_sales_data.csv** and **clean_dataset.csv**
- **generate_raw_data.py**, **build_deck.py**: reproducible build

Data cleaning log

- 12,400 raw rows to 11,996 clean rows
- 400 duplicates; 1,800 sales and 1,800 profit values converted from text
- 3 date formats unified; 1,260 missing cells resolved (0 left)
- 28 quantity entry errors fixed; 430 high-value orders flagged and kept as genuine

Model assumptions (Base)

- Discount cap 20%; marketplace cap 10%; fee cut 1 point
- Retained after cap: 70% (Rec 1), 70% (Rec 2)
- Late deliveries cut 40%; retention +5 points
- Return handling cost 8% of sales
- Costs: $34k one-time, $19k / yr; 36-month horizon, 3-month ramp

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